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DEPARTMENT: COMPUTER ENGINEERING

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Mini Project 1.1: Link Budgets and Coverage Range

1. Setup

Free-space path loss, $L = 20 \cdot \log_{10}(d_{\text{km}}) + 20 \cdot \log_{10}(f_{\text{MHz}}) + 32.44$, was combined with the link budget $P_{\text{RX}} = P_{\text{TX}} + G_{\text{TX}} - L_{\text{path}} + G_{\text{RX}} - L_{\text{other}}$ in Python (NumPy/Matplotlib). Both bands used identical parameters, so any difference comes only from frequency: $P_{\text{TX}} = 40$ dBm, $G_{\text{TX}} = 15$ dBi, $G_{\text{RX}} = 0$ dBi, $L_{\text{other}} = 3$ dB, and a receiver sensitivity of -100 dBm.

Band	Path loss at 0.5 km	P_RX at 0.5 km	Link margin at 0.5 km	Max range ($P_{\text{RX}} > -100$ dBm)
700 MHz (low-band)	83.3 dB	-31.3 dBm	68.7 dB	1358 km
28 GHz (mmWave)	115.4 dB	-63.4 dBm	36.6 dB	34 km

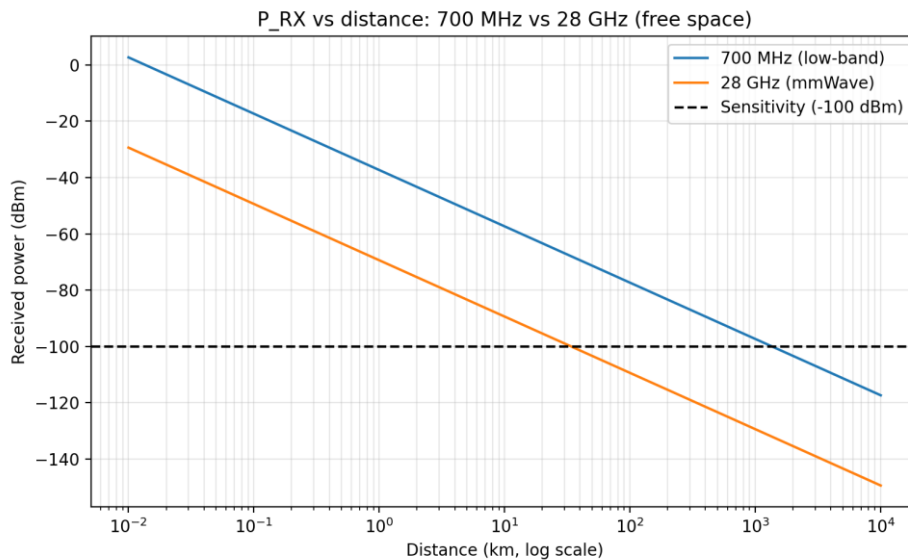


Figure 1: Received power versus distance for 700 MHz and 28 GHz, with the -100 dBm sensitivity line.

2. Analysis

Why the ranges differ. In the path loss formula, frequency enters only through $20 \cdot \log_{10}(f)$. At any fixed distance, the 28 GHz link therefore loses $20 \cdot \log_{10}(28000/700) = 20 \cdot \log_{10}(40) \approx 32$ dB more than the 700 MHz link, which matches the simulated gap (-63.4 dBm versus -31.3 dBm at 0.5 km). The two curves are parallel straight lines on the log-distance axis with a slope of -20 dB per decade, which confirms the code is correct. Solving $P_{RX} = \text{sensitivity}$ for distance gives $d \propto 1/f$, so the range ratio equals the frequency ratio: $1358 \text{ km} / 34 \text{ km} = 40$. This is the coverage-versus-capacity trade-off of Section 1.5: low-band spectrum reaches far, while mmWave offers wide bandwidth but needs dense cells.

Limits of the model. These ranges are free-space figures and are far larger than real deployments. The manual expects mmWave to reach only a few hundred metres. Free space ignores rain absorption, blockage, foliage, building penetration and the radio horizon, all of which hit mmWave hardest, so the absolute ranges should not be read as coverage claims. The relative comparison between bands is the valid result.

3. Lab task results

- **Task 1 (24 dBi at 28 GHz):** the extra 9 dB of gain multiplies range by $10^{(9/20)} = 2.82\times$, from 34 km to 96 km. This recovers part of the gap but not all of it.
- **Task 2 (3.5 GHz):** the range is 272 km, between the other two. It is five times shorter than 700 MHz, matching the $5\times$ frequency ratio.
- **Task 3 (8 dB shadowing margin):** every band shrinks to 40% of its original range (for example 34 km to 13.5 km), because the same dB loss maps to the same distance ratio at any frequency.
- **Task 4 (SNR and capacity, B = 10 MHz, noise -104 dBm):** P_{RX} falls 20 dB per decade of distance, but capacity falls slowly because of the logarithm in $C = B \cdot \log_2(1 + \text{SNR})$. For 700 MHz, capacity drops from 288 Mbps at 0.1 km to 155 Mbps at 10 km, even though P_{RX} falls by 40 dB. Using 10 MHz for both bands understates mmWave, which normally has hundreds of MHz.
- **Task 5 (double power versus double bandwidth):** at 0.5 km on 700 MHz, doubling bandwidth raises capacity $\times 1.92$, while doubling power raises it only $\times 1.04$. Bandwidth

scales capacity linearly, but power enters through $\log_2(1 + \text{SNR})$, so a 3 dB gain adds only a fraction of a bit per second per hertz.

- **Task 6 (TDD versus FDD duty cycle):** a link budget assumes continuous transmission, which holds for FDD. A TDD transmitter is on only during its own direction's slots, and guard periods add dead time, so its duty cycle is about $(\text{DL slots} \div \text{total slots}) \times (1 - \text{guard fraction})$. The instantaneous link budget is unchanged, but average throughput is lower and average transmit power is reduced.